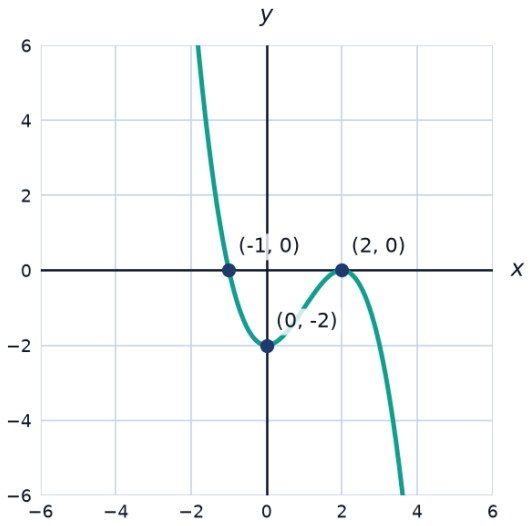
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of a third-degree polynomial function $P(x)$ is shown in the standard (x, y) coordinate plane below. The graph crosses the x-axis at $(-1, 0)$, is tangent to the x-axis at $(2, 0)$, and passes through the point $(0, -2)$. What is the value of $P(-2)$?



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If the real numbers x and y satisfy the system of equations $x^3 - 3xy^2, = 10; 3x^2y - y^3, = 5$ what is the value of $x^2 + y^2$?

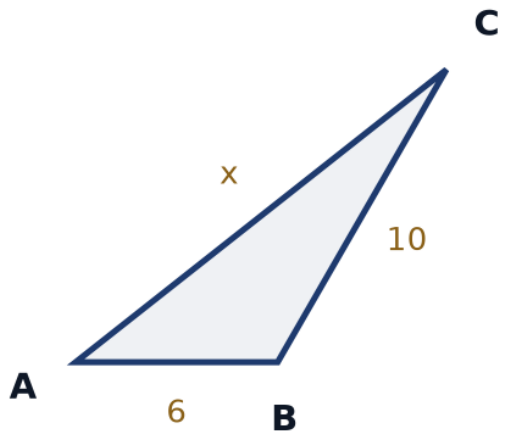
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the equations of the asymptotes of the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$.

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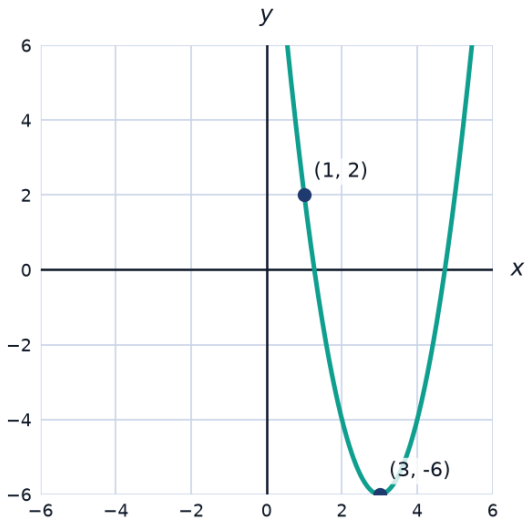
4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC shown below, side AB has length 6, side BC has length 10, and the area of the triangle is $15\sqrt{3}$. If $\angle B$ is an obtuse angle, what is the length of side AC, denoted by x?



5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

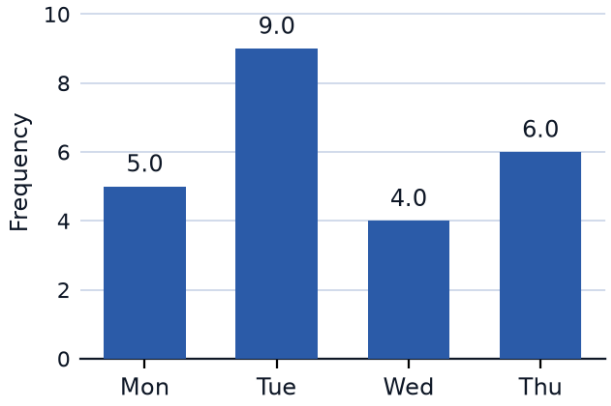
The quadratic function $f(x) = ax^2 + bx + c$ is graphed in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(3, -6)$, and the graph passes through the point $(1, 2)$. What is the value of b?



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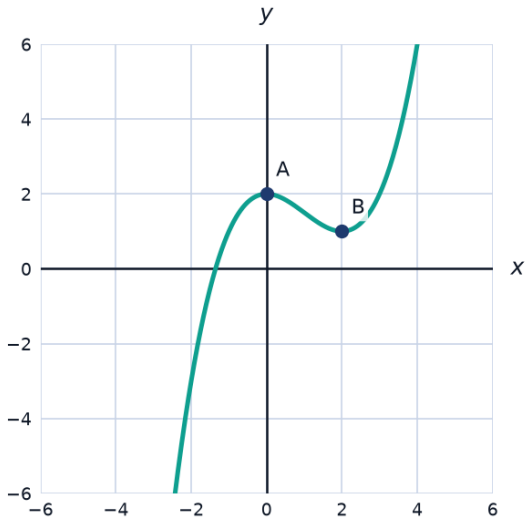
6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The bar chart below displays the number of cars sold by an auto dealership each day from Monday through Thursday. What was the mean number of cars sold per day over these 4 days?



7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the horizontal asymptote of $f(x) = \frac{3x^2}{x^2 + 1}$?

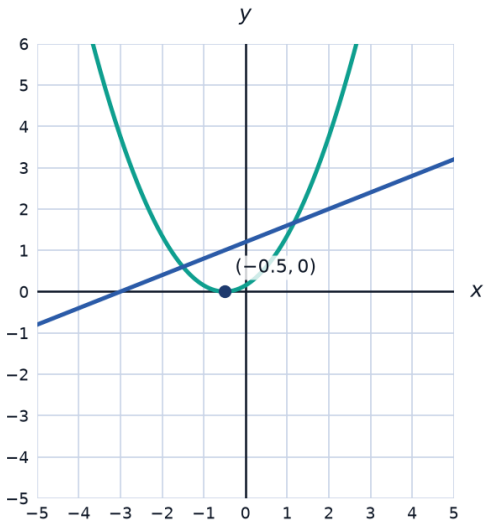


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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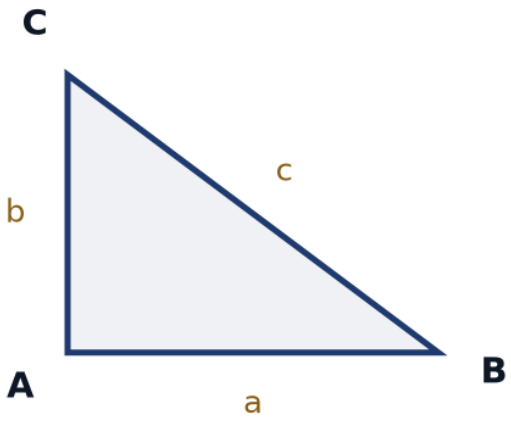
8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For what values of k does $x^2 - kx + 9 = 0$ have equal real roots?



9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

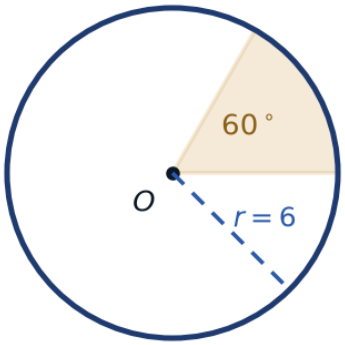
In triangle ABC, $a = 8$, $A = 30^\circ$, $B = 45^\circ$. Find b.



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10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the exact value of $\sin 135^\circ$.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In the standard (x, y) coordinate plane, a right triangle has vertices at $A(-3, -1)$, $B(5, 5)$, and $C(4, k)$, where the right angle is at vertex C and $k > 0$. What is the value of k?

12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In the figure below, ΔABC is a right triangle with right angle at B, and $\overline{BD}$ is an altitude to hypotenuse $\overline{AC}$ with length h. If $AD = 9$ and $DC = 16$, what is the perimeter of ΔABC?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the area enclosed between $y = x^2$ and $y = x + 2$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the system: $x^2 - y = 3$ and $x + y = 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Advanced Functions

Express $0.\overline{27}$ as an irreducible fraction.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Trigonometry

Find the inverse function of $f(x) = \frac{2x-1}{x+3}$.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find all real solutions to the rational equation $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Let matrices $A = (3, -1; 2, 4)$ and $B = (1, 5; 0, -2)$. Compute AB, $\det(A)$, $\det(B)$, and verify that $\det(AB) = \det(A)\det(B)$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Advanced Plane Geometry

Solve for x: $3^{x+1} = 5^x$. Express your solution in terms of natural logarithms.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Statistics, Counting & Probability

When the polynomial $P(x) = 2x^3 - kx^2 + 5x - 6$ is divided by $x - 2$, the remainder is 8. What is the value of the constant k?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When the polynomial $P(x) = 2x^3 - 5x^2 + kx - 8$ is divided by $(x - 3)$, the remainder is 10. What is the value of k?

a. 3

b. 5

c. 7

d. 9

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the quotient when $4x^3 + 2x^2 - 6x + 5$ is divided by $(2x + 1)$?

a. $2x^2 - 3$

b. $2x^2 + x - 3$

c. $2x^2 - 3x + 1$

d. $2x^2 + 3$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $(x + 2)$ and $(x - 1)$ are both factors of $x^3 + ax^2 + bx - 6$, what is the value of $a + b$?

a. -1

b. 1

c. 3

d. 5

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let r and s be the roots of $2x^2 - 6x + 1 = 0$. What is the value of $r^2 + s^2$?

a. 7

b. 8

c. 9

d. 10

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If the roots of $x^2 - kx + 36 = 0$ are in the ratio $1 : 4$, what is the positive value of k?

a. 12

b. 15

c. 18

d. 20

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive y-value of the points of intersection of $x^2 + y^2 = 25$ and $y = x^2 - 5$.

a. 0

b. 3

c. 4

d. 5

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many real solutions (x, y) satisfy the system $xy = 6$ and $x^2 + y^2 = 13$?

a. 1

b. 2

c. 4

d. 0

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the system $y = 2x + k$ and $y = -x^2 + 4x - 1$, for what value of k are the lines tangent?

a. 0

b. 1

c. 2

d. 3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of a will the system $x + y = 3$, $2x + 2y = a$ have no solution?

a. Any $a \neq 6$

b. $a = 6$

c. $a = 0$

d. $a = 3$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the value of x in the unique solution to the matrix equation $[[2, 1], [3, 4]][[x], [y]] = [[7], [18]]$?

a. 1

b. 2

c. 3

d. 4

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the system of linear equations below: $\{3x - 2y = 8; kx + 6y = 5$ For which of the following values of the constant k does the system have no solution (x, y)?

a. -9

b. -4

c. 4

d. 9

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the system of linear equations below, k is a real constant. $(k - 1)x + 6y, = 18; 2x + ky, = 12$ For which value of k does the system have infinitely many solutions (x, y)?

a. -4

b. -3

c. 3

d. 4

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the standard (x, y) coordinate plane, the system of equations $\{y = |x^2 - 4|; y = 2x + k$ has exactly 3 distinct real solutions (x, y). What is the sum of all possible values of the real constant k?

a. 4

b. 5

c. 9

d. 11

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34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the standard (x, y) coordinate plane, the lines with equations $2x - y = 7$ and $x + y = 5$ intersect at the point (a, b). What is the value of a?

a. 1

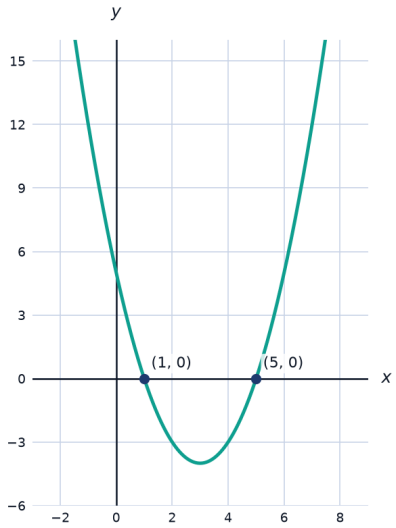
b. 3

c. 4

d. 5

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown in the standard (x, y) coordinate plane below. A horizontal line $y = k$ intersects the graph of $f(x)$ at two points that are 8 units apart. What is the value of k?



a. 12

b. 8

c. 4

d. −3

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In the standard (x, y) coordinate plane, the line $y = 2x - 3$ intersects the parabola $y = x^2 - 6x + c$ at two distinct points, P and Q. If the distance between P and Q is $4\sqrt{5}$, what is the value of the constant c?

a. −7

b. 9

c. 12

d. 15

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In the standard (x, y) coordinate plane, at what values of x do the graphs of $y = x^2 - 3x$ and $y = 2x - 4$ intersect?

a. $x = 1$ and $x = 4$

b. $x = -4$ and $x = -1$

c. $x = -2$ and $x = 2$

d. $x = 2$ and $x = 3$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

The line $y = 1x - 1$ intersects the parabola $y = x^2 + 1x - 2$ at two points. What is the sum of the x-coordinates of the intersection points?

a. 2

b. 1

c. -1

d. 0

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Advanced Plane Geometry

Solve the system $2x + 5y = -45$ and $5x - 3y = -4$. What is $x + y$?

a. -5

b. -12

c. -11

d. -13

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Statistics, Counting & Probability

Given $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 3 \rangle$, what is the cross product $\mathbf{u} \times \mathbf{v}$?

a. $\langle 2, 3, 0 \rangle$

b. $\langle 6, 3, 1 \rangle$

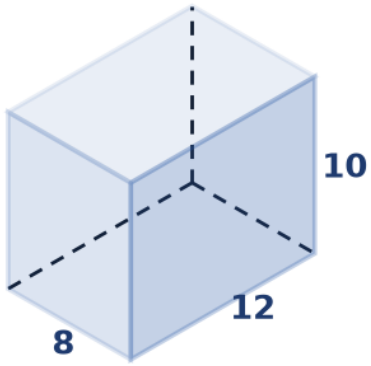
c. $\langle 6, -3, 1 \rangle$

d. $\langle -6, -3, -1 \rangle$

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Advanced Number Theory

Find the volume of a rectangular prism with length 8, width 12, and height 10.



a. 960

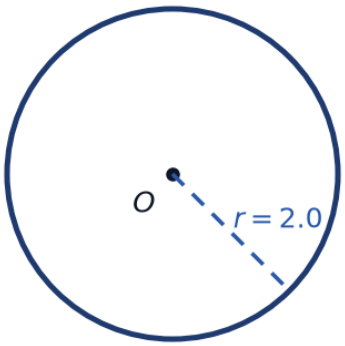
b. 968

c. 592

d. 948

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Advanced Functions

What is the radius of the circle $(x - 8)^2 + (y - 4)^2 = 4$?



a. 1

b. 2

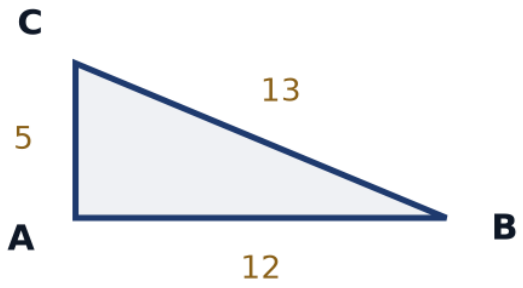
c. 4

d. 3

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Advanced Trigonometry

A right triangle has legs 5 and 12 with hypotenuse 13. Find $\tan\theta$, where θ is the angle opposite the side of length 5.



a. $\frac{12}{13}$

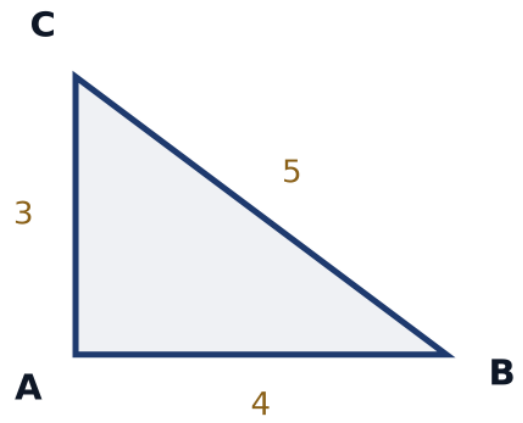
b. $\frac{13}{5}$

c. $\frac{12}{5}$

d. $\frac{5}{12}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Pre-Calculus Concepts

In right triangle: opp=4, hyp=5. Find $\sin\theta$.



a. $\frac{3}{5}$

b. $\frac{4}{5}$

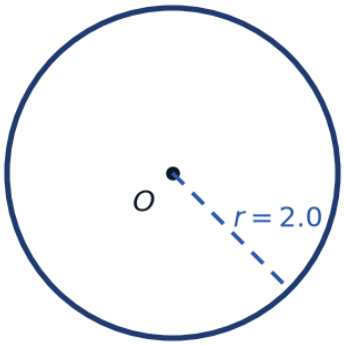
c. $\frac{4}{3}$

d. $\frac{3}{4}$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Find the circumference of a circle with radius 2. (Leave your answer in terms of π.)



a. 2π

b. 8π

c. 6π

d. 4π

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